

Stratified Distributed SGD for Matrix Factorization | By Yagel Alfasi

Based on the paper: “Large-Scale Matrix Factorization with Distributed Stochastic Gradient Descent” by R. Gemulla, E. Nijkamp, P. J. Haas & Y. Sismanis, KDD 2011.

Course: Distributed Optimization and Multiagent Systems

Introduction

DSGD factorizes a sparse matrix $V \approx WH$ by minimizing a loss that decomposes over the observed entries, $L(W, H) = \sum_{(i,j) \in \mathcal{Z}} l(V_{ij}, W_{i*}, H_{*j})$, where an SGD step on (i, j) touches only row i of W and column j of H . Naive parallelism fails because two nodes drawing entries from a shared row or column overwrite each other. The paper’s fix is structural: two points are *interchangeable* when

$$i_1 \neq i_2 \quad \text{and} \quad j_1 \neq j_2, \quad (1)$$

and Theorem 2 states that interchangeable points may be applied in any order, or simultaneously, with a result identical to some sequential run. The guarantee concerns *which coordinates a step touches*, not the loss surface - what separates DSGD from Hogwild-style arguments.

The Algorithmic Idea: Interchangeability and Strata

Partition rows and columns into d groups each. A *stratum* is a set of d blocks sharing no row group and no column group (a d -monomial); its blocks are mutually interchangeable and run on d nodes with no coordination during the step (Figure 1).

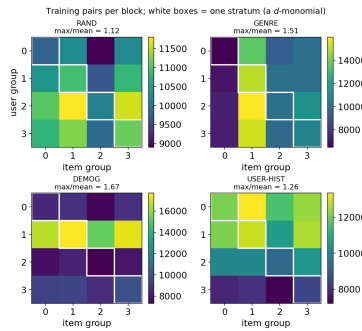


Figure 1. Training pairs per block on the $d=4$ grid; white boxes mark one stratum - d blocks sharing no row or column group, touching disjoint rows of W and columns of H , so each node runs one block with no coordination.

A subepoch processes one stratum, an epoch cycles through d of them. Data is partitioned once and stays put; only factor blocks, $O((m+n)k)$, cross the network. SSGD writes $L = \sum_s w_s L_s$ with $w_s = |\mathcal{Z}_s|/|\mathcal{Z}|$ and steps on one stratum at a time; DSGD’s Theorem 3 shows the parallel execution equals *some* valid sequential SSGD run, so its convergence transfers. The loss need only decompose over entries, so squared loss, KL-divergence and regularized variants are covered where ALS must be re-derived per objective; a “bold driver” step size (+5% after a falling epoch, halved after a rising one) removes hand-tuned schedules. DSGD out-converges ALS, PSGD and DGD on Netflix and scales to 25.6B ratings; the authors’ own caveat is that past some cluster size, communication sets the ceiling.

Connection to the Course

Which framework. This is distributed *optimization* of one shared objective: no node has a private payoff and none chooses strategically, so there is no equilibrium to characterize. Forced into a game it is the identical-interest case - an exact potential game whose potential is L itself - so the global optimum is always an equilibrium; were L convex the pure price of anarchy would be 1, and any gap between equilibria can only come from the bilinearity of L , never from conflicting interests.

How the assumptions differ. In the algorithms studied in class, every agent holds a local copy x_i of the shared variable, and the analysis centers on how far these copies drift apart: the disagreement error $\phi(\mathbf{x}) = \|(I - W)\mathbf{x}\|^2$, bounded through the spectral gap of a doubly stochastic W . DSGD has no such term. Its nodes hold *disjoint coordinates* of one parameter vector and (1) guarantees exact reassembly, so $\phi \equiv 0$ by construction - no spectral gap or communication graph appears in its rate; consensus is not achieved by averaging but made unnecessary by the partition, at the price of an orchestrator. The second difference is the one measured here: stochastic distributed optimization assumes unbiased local gradients and bounds heterogeneity by $\frac{1}{N} \sum_i \|\nabla f_i(\bar{\mathbf{x}}) - \nabla \Phi(\bar{\mathbf{x}})\|^2 \leq \sigma^2$. Stratified sampling violates the first directly,

$$\mathbb{E}[g_n | \mathcal{F}_{n-1}] = \nabla L_{s(n)}(\theta_{n-1}) \neq \nabla L(\theta_{n-1}), \quad (2)$$

and semantic partitions deliberately make σ^2 larger: grouping similar items or users together creates exactly the kind of heterogeneity this bound is meant to keep small. Unbiasedness returns only in time average, with strata visited at frequencies w_s - the paper’s regenerative-process argument, with the epoch as the cycle.

Which course tools carry over. The class proof of almost-sure convergence runs through the filtration $\mathcal{F}_t$ and the Robbins-Siegmund theorem. The theorem itself still applies here, but (2) breaks the step that feeds it. In the class recursion for distributed SGD, the cross term $-2\eta(t)\langle \mathbb{E}\{\mathbf{g}_s(t) \mid \mathcal{F}_t\} - \nabla\Phi(\bar{\mathbf{X}}(t)), \bar{\mathbf{X}}(t) - \mathbf{x}^* \rangle$ shrinks at rate $O(\eta(t))$ because the local gradients are unbiased, and this is exactly what puts the recursion into Robbins-Siegmund form. Under stratified sampling this term is $O(1)$: it does not shrink with the step size and its sign is not controlled, so it cannot be pushed into the summable perturbation ξ_t . Figure A2 checks the consequence empirically: if the accumulated stratum noise grows like $\sqrt{n}$ the increments still behave like a martingale, while growth like n marks a real drift.

What the Paper Leaves Open, and the Direction I Propose

Three gaps motivate the direction I take: the convergence result constrains the stratum *sequence* only through w_s ; the grid is assumed random, and the paper never asks what happens when the partition is dictated by the data; and interchangeability is stated for models whose row factor is private to its row, saying nothing about models sharing parameters across rows - most modern recommenders. This leads to four hypotheses (H1–H4), tested in the simulation: **H1** (load): semantic item blocks are load-imbalanced, inflating makespan; **H2** (bias): semantic strata are gradient-biased in the sense of (2), removable by w_s weighting; **H3** (shared factors): a history-encoder shares row factors, so DSGD is only approximate and the partition controls how badly; **H4** (explanations): SPINRec attributions read out the learned item-item geometry, so the blocking scheme should move explanation fidelity, not only accuracy.

Simulation

1. Setup and metrics

Data and model. MovieLens-1M binarized at rating ≥ 4 ; the 1,500 most-rated movies and 2,000 sampled users with ≥ 10 positives, giving 168,980 training interactions at 5.6% density, split leave-one-out by timestamp. The recommender is the history-encoder used in my explainability work,

$$s(u, j) = \sigma((A^\top x_u)^\top (B^\top e_j)) = \sigma\left(\left(\sum_{i \in \text{hist}(u)} A_{i*}\right)^\top B_{j*}\right), \quad (3)$$

with $x_u \in \{0, 1\}^I$ the history vector. The item factor B_{j*} is private to item j , but the user factor is a sum of **item-indexed rows of A shared across users**, so (1) fails: the case DSGD’s guarantee does not cover. The tension is explicit: gradient-based explanation methods need the score differentiable in the user’s history items, which classic MF’s free per-user embedding does not provide; the structure creating that path - shared rows of A in place of a private user row - is exactly what removes the ownership DSGD requires. This explainability and DSGD’s exactness trade against each other by construction. Biases are dropped (every example touches them, making all pairs conflict trivially); negatives are drawn within the block’s item shard, since a node holds only that shard. Training settings shared by all arms are listed in Table 1. An epoch visits the d strata in a fixed rotation: in subepoch t , node r processes the block of row group r and column group $(r + t) \bmod d$, so every block combination is covered exactly once per epoch in a deterministic order. The fixed order rules out the schedule itself as a source of drift.

k	loss	mini-batch	step size	blocks d	epochs	seeds
32	BCE, 4 negatives	256	fixed	4 (2, 8 in scaling)	20	5

Table 1: Training settings, identical across all arms.

Arms. **RAND** random grid re-drawn each epoch (the paper’s own design partitions once); **FIXED** random but fixed, isolating re-permutation; **GENRE/GENRE-BAL** items by k -means on genre indicators, the latter rebalanced to equal size; **YEAR** items by release-year quartiles; **DEMOG** users by age bracket; **USER-HIST** users by k -means on history vectors; **SEQ** centralized. Every stratum activates all d user groups at once, and the rows of A a block touches are $\bigcup_{u \in \text{row group}} \text{hist}(u)$: which rows are contested is fixed by the user partition alone, item-side schemes change only magnitudes and load.

Measurements. *Contested mass*: per row of A , summed block-delta magnitude minus the largest single block’s share (the plain fraction of contested rows saturates near 1). *Theorem-2 gap*: here θ_0 is the parameter vector at the start of a subepoch, θ_{par} is the result of running the subepoch’s d blocks in parallel and merging their updates, and θ_{seq} is the result of running the same blocks one after another. The gap $\|\theta_{\text{par}} - \theta_{\text{seq}}\|/\|\theta_{\text{par}} - \theta_0\|$ measures how far the merged result is from the sequential one, relative to how far the subepoch moved the parameters at all; when Theorem 2 holds it is exactly zero. It is reported per local-step count, so the collision effect is not confounded with how far each block travels. *Fidelity*: SPINRec [2] - stochastic path integration, integrated gradients from a randomly drawn baseline user. For this bilinear model $\partial s/\partial x = \sigma'(z)(AB_{j*})$ and σ' is a positive scalar that factors out of

the ranking, so the attribution is closed-form: a read-out of $AB^\top$ over the user’s history (verified against autograd). Masking the top- k attributed items I report $\text{CDCG}@k = 1/\log_2(\text{rank} + 1)$ with the target ranked among all items; **lower is better** (full curves over k in Figure A1). All arms share the same 800 explained users, baseline draws and initialization.

Communication. In classic MF each node owns one block of W and one of H , so a subepoch ships only those blocks, $(m + n)k/d$ floats, and an epoch costs $d(m + n)k$. Here A is shared: a node’s updates touch rows of A across *all* items, so it must ship the full update ΔA (Ik floats) plus its own block of B (Ik/d), and an epoch costs $d(d + 1)Ik - (d + 1)I/(m + n)$ times classic MF, $2.14\times$ at $d = 4$ and growing linearly with d . The overhead comes from the model sharing A , not from the partition choice. At our scale the model traffic per epoch (3.84 MB) already exceeds the entire training data (1.35 MB), so shipping parameters instead of data pays only when $|\mathcal{Z}| \gg d(d + 1)Ik$ - the web-scale regime the paper targets, not the one simulated here.

2. Does Theorem 2 survive a shared row factor?

Scheme	Gini	straggler max/mean	contested mass	Thm.2 gap (100 steps)	val HR@10
RAND	0.053	1.11	0.678	0.123	0.0755±0.0086
FIXED	0.048	1.11	0.672	0.123	0.0793±0.0036
GENRE	0.155	1.42	0.658	0.143	0.0714±0.0031
GENRE-BAL	0.038	1.09	0.666	0.162	0.0699±0.0028
YEAR	0.074	1.15	0.663	0.151	0.0746±0.0041
DEMOG	0.173	1.58	0.639	0.127	0.0870±0.0059
USER-HIST	0.104	1.22	0.594	0.323	0.0775±0.0084
SEQ (centralized)	-	-	-	-	0.0880±0.0025
CLASSIC-MF control	-	-	-	2.1e-18	-

Table 2: Structural diagnostics and final accuracy, mean $\pm$ spread over 5 seeds. **RAND** and **FIXED** differ only in whether the grid is re-drawn each epoch.

On classic free-embedding MF the parallel merge reproduces sequential execution to a relative $2 \cdot 10^{-18}$ at every local-step count. For (3) the deviation ranges from $1.7 \cdot 10^{-2}$ (**RAND**) to $8.3 \cdot 10^{-2}$ (**USER-HIST**) after one local step and reaches 0.12-0.32 after 100. The ordering is the surprise: **USER-HIST** has the *lowest* contested mass (0.594 vs 0.678) and the *largest* gap (0.323 vs 0.123). No contradiction - interchangeability fails through a second-order term, so the error is the *product* of how far each block moved a shared row, not the count of shared rows: clustering similar users contests fewer rows but concentrates each block’s updates onto them, and the products grow faster than the count falls. For a sharding policy this inverts the naive advice: with many local steps between merges, blocks should be internally *heterogeneous*. Scaling to $d \in \{2, 4, 8\}$ (Table A1), accuracy falls monotonically for every scheme - **RAND** loses 37% of HR@10 from $d=2$ to $d=8$ - while the gap is not monotone: it grows for random and genre grids but collapses for **USER-HIST** at $d=8$ ($0.323 \rightarrow 0.113$), plausibly because eight history-clusters stop overlapping in item support - a conjecture this experiment does not test.

3. Load, accuracy and the drift diagnostic

Load (hypothesis H1). The whole spread across schemes is 0.017 HR@10 and the best distributed arm lands within 0.001 of centralized, so no partition choice here is catastrophic. **DEMOG** and **GENRE** carry the expected straggler cost (1.58 and 1.42 vs 1.11) and **GENRE-BAL** removes it (1.09) for 0.002 HR@10, inside the seed spread: load imbalance is a scheduling problem, not a statistical one. Re-permutation costs something - **RAND** 0.0755 vs **FIXED** 0.0793, two arms identical in every structural diagnostic - but the effect shrinks with training: **FIXED** closes 53% of **RAND**’s gap to centralized at epoch 10 and only 30% by epoch 20 (Figure A4). It is mainly a *convergence-rate* penalty, not an asymptotic one.

Drift (hypothesis H2) fails in the opposite direction to the one I assumed: I expected semantically homogeneous strata to be the biased ones, but the accumulated projected noise grows close to linearly for the *random* grids, reaching 1.40 and 1.29 times the $c\sqrt{n}$ scale for **RAND** and **FIXED**, against 0.567 for **GENRE** (Figure A2). Every curve stays inside the Azuma envelope; the shape is what separates them. Applying the SSGD weights $w_s \propto |\mathcal{Z}_s|$ moves **GENRE** only from 0.567 to 0.515, identifying what the bias is *not*: were it the stratum-weighting bias w_s corrects, weighting would have removed it. What remains is the merge itself - summing d block updates that overlap on shared rows over-counts them, and with a re-drawn grid the overlap pattern is fresh every epoch, so the over-count never settles into anything the model can compensate for. (Seed 0 only; magnitudes indicative.)

Scheme	CDCG@5	POS@10 at $k=5$	$\Delta\text{CDCG}@5$ vs RAND
SEQ	0.3810	0.597	-
RAND	0.4565	0.644	-
FIXED	0.4859	0.677	+0.0294
GENRE	0.3950	0.601	-0.0615
GENRE-BAL	0.4079	0.623	-0.0486
YEAR	0.4220	0.608	-0.0345
DEMOG	0.4158	0.654	-0.0407
USER-HIST	0.4597	0.688	+0.0032

Table 3: SPINRec deletion fidelity at $k=5$, averaged over the shared 800 explained users and 5 seeds. CDCG@5 scores the target’s rank after masking the user’s 5 most-attributed history items: a faithful explanation drops the target far, so lower is better. POS@10 is the probability the target survives the top 10 after the same masking, lower is better (full curves in Figure A5). Δ CDCG@5 is the difference vs RAND; negative means more faithful than a random grid. Significance is discussed in the text.

4. Explanation fidelity

Hypothesis H4: right direction, insufficient evidence. Every item-semantic scheme sits below RAND in CDCG@5, and GENRE comes closest of any distributed arm to centralized - the ordering H4 predicted. Throughout, p is the p -value of a paired t -test on the CDCG@5 difference vs RAND: the probability of a difference at least this large arising by chance if the schemes did not truly differ; small p means the difference is unlikely to be noise. But the statistics turn on *what is replicated*. Pairing per user gives 800 observations per seed and small p -values - GENRE at $p \leq 0.022$ in every seed - yet those 800 users all explain the same *pair of trained models*, so the test asks whether these two models differ, not whether the schemes do. The unit carrying the scheme effect is the seed, where the picture is weaker: GENRE’s per-seed differences are $-0.064, -0.054, -0.113, -0.111, +0.035$, four of five negative, seed-level $p = 0.084$. USER-HIST makes the point in reverse: per-user $p = 0.003$ on a mean difference of $+0.003$, significant only because 800 paired users resolve almost anything. Honest summary: item-side blocking moves fidelity in the predicted direction in four of five model instances, at $p \approx 0.08$ that would not survive correction for six comparisons. Caveats: the arms do not reach equal accuracy, so the CDCG ordering partly tracks HR@10; and neither measure of the interchangeability violation predicts the fidelity shift (Figure A3). One endpoint is cleaner: attributions correlate with the centralized model’s at $\rho = 0.46$ (DEMOG) and 0.42 (USER-HIST) against 0.28 (RAND), with DEMOG the only seed-level $p < 0.05$ (0.042 , uncorrected). User-side schemes give the most centralized-like explanations despite USER-HIST’s worst parameter gap - by (3) the explanation depends on $AB^\top$, and two distant parameter vectors can induce nearly the same item-item geometry.

5. What disjointness is worth

Letting a node sample negatives globally, and so write to item factors it does not own, is the one manipulation that breaks DSGD’s premise outright. Accuracy falls moderately, $0.0765 \rightarrow 0.0620$; the explanations collapse, CDCG@5 $0.350 \rightarrow 0.925$, near the ceiling. Every other arm varies *how* the grid is drawn while keeping blocks disjoint and moves accuracy by a few points; abandoning disjointness is categorically worse. This connects back to the course comparison: the algorithms from class pay for agreement with a consensus step and carry the disagreement term ϕ , while DSGD pays with the partition itself. Once blocks are no longer disjoint, nothing in the method controls the resulting error. (Single seed; treat magnitudes as indicative.)

6. Conclusions

Bottom line. The decisive result is not any hypothesis but disjointness itself: breaking it is the single largest effect measured - accuracy drops moderately ($0.0765 \rightarrow 0.0620$) and the explanations collapse (CDCG@5 $0.350 \rightarrow 0.925$). Disjointness is what DSGD’s guarantee actually buys. The four hypotheses resolve: **H1** (load): confirmed but harmless - semantic blocks do carry a straggler cost (1.58 and 1.42 vs 1.11), yet balancing removes it for 0.002 HR@10 and the whole accuracy spread across schemes is 0.017 , so load imbalance is a scheduling problem, not a statistical one. **H2** (bias): refuted in direction - the drift belongs to the *random* grids, not the semantic ones, and the w_s weighting barely touches it, so the bias is the merge of overlapping updates, not the stratum-weighting bias the theory corrects. **H3** (shared factors): confirmed and sharpened - the parallel-vs-sequential gap is real (0.12 - 0.32) and governed by how concentrated each block’s updates are on the shared rows, not by how many rows are shared, so blocks should be internally heterogeneous. **H4** (explanations): right direction, insufficient evidence - item-side blocking improves fidelity in four of five seeds, but at $p \approx 0.08$ this is grounds for a larger study, not a claim. In one line: keep the blocks disjoint and DSGD tolerates any partition rule tested at a cost of a few accuracy points; give up disjointness and nothing in the method controls the error - the paper’s own point, restated from the other side.

Limitations and Further Work

The simulation executes blocks sequentially from shared starting parameters, reproducing the arithmetic of parallel execution exactly but saying nothing about communication - where the paper’s own experiments found the real ceiling; the straggler ratio is a makespan *model*, not a measurement. The martingale increment is evaluated once per subepoch and tracked for A only; all fidelity numbers are at $d=4$, and the explained cohort excludes users with fewer than 20 history items. Three directions follow: whether the re-permutation penalty vanishes asymptotically is what the course’s rate machinery answers; if deviation is driven by concentration, the partition objective should minimize the largest share any single parameter receives under a load constraint; and stratum scheduling from staleness and recent loss, as a restless bandit, could be tested against the paper’s fixed round-robin.

References

- [1] R. Gemulla, E. Nijkamp, P. J. Haas, Y. Sismanis. *Large-Scale Matrix Factorization with Distributed Stochastic Gradient Descent*. KDD 2011.
- [2] O. Barkan, Y. Schein, Y. Elisha, V. Bogina, M. Baklanov, N. Koenigstein. *Fidelity-Aware Recommendation Explanations via Stochastic Path Integration*. AAAI 40(17), 14484-14492, 2026.
- [3] F. M. Harper, J. A. Konstan. *The MovieLens Datasets: History and Context*. ACM TiiS 5(4), 2015.

A Supplementary tables and figures

Scheme	d	Thm. 2 gap	val HR@10
FIXED	2	0.078	0.0920
FIXED	4	0.123	0.0803
FIXED	8	0.132	0.0712
GENRE	2	0.087	0.0897
GENRE	4	0.143	0.0732
GENRE	8	0.125	0.0705
RAND	2	0.078	0.0943
RAND	4	0.123	0.0737
RAND	8	0.132	0.0592
USER-HIST	2	0.317	0.0913
USER-HIST	4	0.323	0.0793
USER-HIST	8	0.113	0.0755

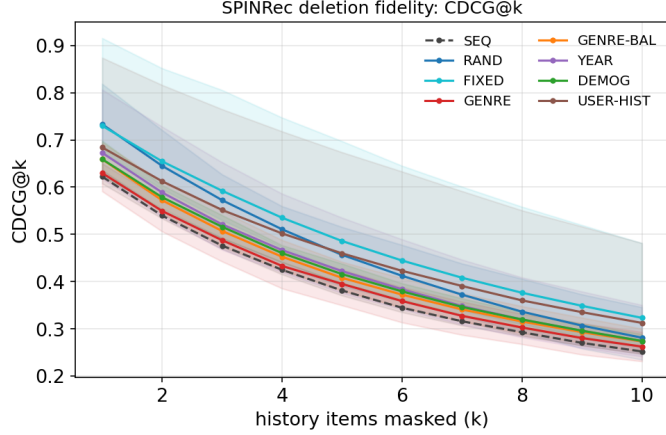


Table A1 (left). Scaling in the block count d : the Theorem-2 gap after 100 local steps and the final validation accuracy; **DEMOG** is defined for $d=4$ only. Mean over three seeds.

Figure A1 (right). SPINRec deletion fidelity, CDCG@k versus number of masked history items; lower is better.

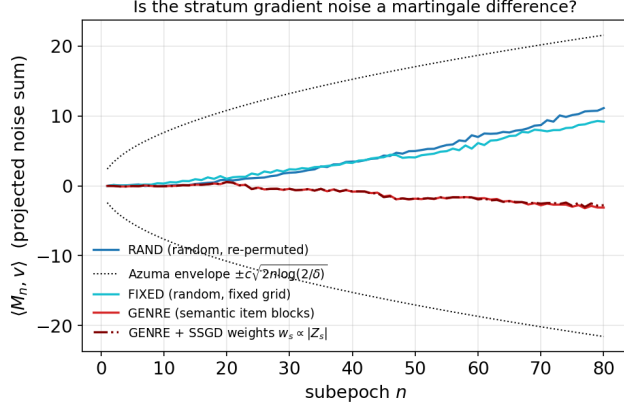
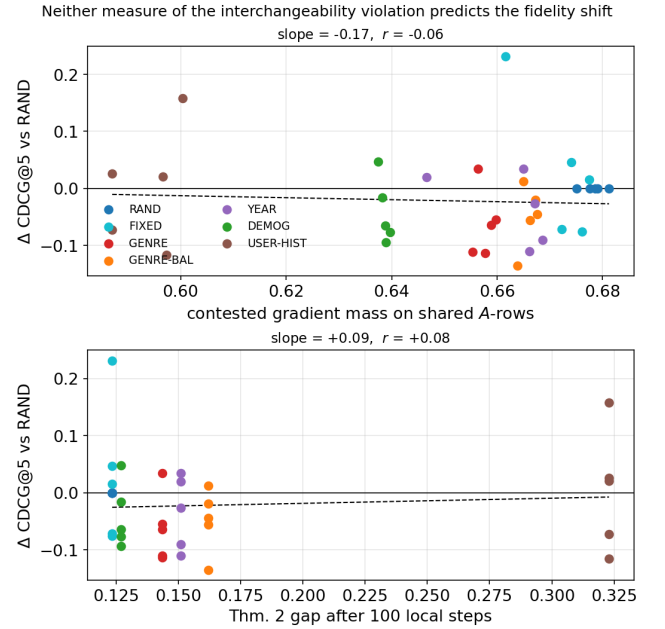


Figure A2 (left). Empirical test of the bias in Eq. (2): accumulated projected gradient noise per subepoch against an Azuma envelope. Growth like $\sqrt{n}$ means the increments still behave like a martingale, as the Robbins-Siegmund argument needs; the near-linear growth of the random grids marks a systematic drift. Figure A3 (right). Neither measure of the interchangeability violation predicts the paired fidelity shift.



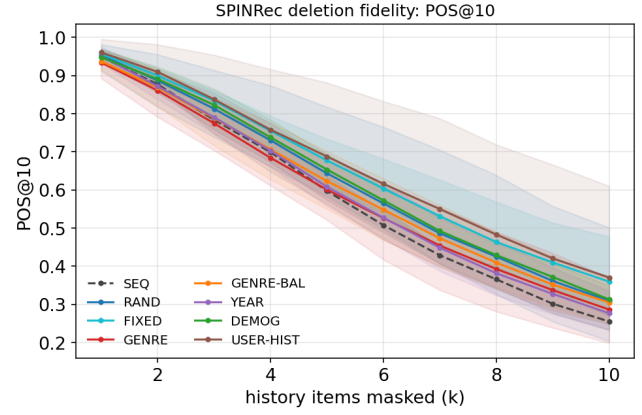
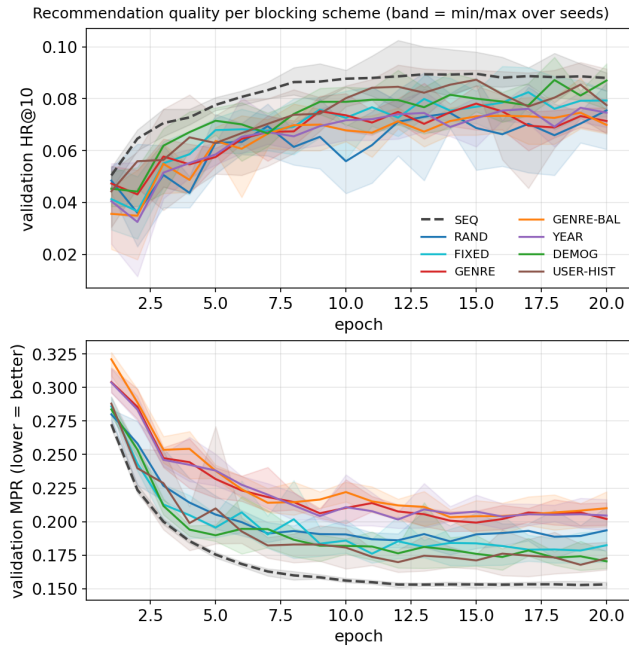


Figure A4 (left). Validation HR@10 and MPR per epoch, band = min/max over seeds; supports the convergence-rate reading of the *RAND* vs *FIXED* gap. Figure A5 (right). SPINRec POS@10: probability the target survives the top 10 after masking its k top-attributed history items.